AI-Powered Model for Evaluating UI/UX Design Based on HCI Principles: Fitts's Law and Miller's Law

BY

Group ID: 13

TG/2020/673 – M.G.H. Nirmani

TG/2020/704 – T. Madhusha

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AI-POWERED MODEL FOR EVALUATING UI/UX DESIGN BASED ON HCI PRINCIPLES: FITTS'S LAW AND MILLER'S LAW

A Proposal Presented

By

GROUP ID: 13

TG/2020/673 – M.G.H. NIRMANI

TG/2020/704 – T. MADHUSHA

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Supervisor: _____

(P. H.P. Nuwan Laksiri)

Co-Supervisor : _____

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1. Introduction

In the modern world, user interface (UI) and user experience (UX) designs play a critical role in software development and web development by determining the effectiveness and usability of different digital products. Poor UI/UX design makes users feel uncomfortable, disengaged, frustrated, leading to lower retention rates and reduced system usability. However, despite the rapid advancement in UI design tools and techniques, ensuring that UI adheres to established Human Computer Interaction (HCI) principles remains a challenge. This research mainly focused on evaluating quantitative HCI principles in UI/UX design using an AI-powered model, with the aim of automating and improving the design evaluation process

In the IT field, especially when considering HCI and UI/UX design principles such as Fitts's Law, Miller's Law are widely used as quantitative measures to indicate usable design. Previous research has explored the importance of these and other principles and has developed some rule based or semi automates tools to design UI/UX design. And most tools depend on manual process and often focus on only one or two principles at a time. However, there is a lack of comprehensive AI-model capable of analyzing and evaluating UI design to automatically verify whether it meets these quantitative principles.

This research aims to develop an AI-powered model that can evaluate UI/UX design using Fitts's Law and Miller's Law. This system will allow developers (designers) to determine in real time whether their UI design meets these principles. By receiving early feedback on the UI design, the development lifecycle can be improved, reducing depend on post deployment feedback.

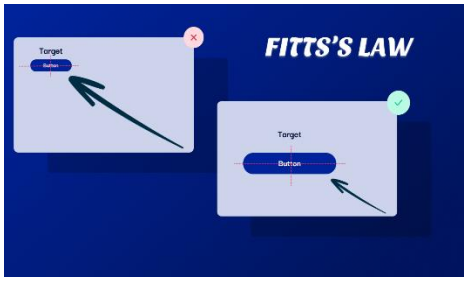

Figure 1.1: Fitts's Law

Figure 1.2: Miller's Law

2. Background

UI/UX design plays a crucial role in the development of digital products as it directly impacts how users interact and engage with them. A well-designed UI can make a digital product visually appealing, intuitive, and easy to navigate [3]. According to the HCI principles, which are the guidelines helped to designers to create UI (user interface), and UX (user experience) that efficient, easily accessible and enjoyable. Also, these HCI principles play crucial role in UI/UX design when providing user satisfaction. When the previous researches are focusing on how the UI/UX principles and guidelines were developed, how it will be worked, how the formulas and equation implement and etc. Moreover, the related researches' methodologies mainly led to the manual approaches and methods. Thus, this research aims to evaluating quantitative HCI principles especially, Fitts's law and Miller's law through an AI-powered model which helps to designers to create and enhance the UI/UX design, user and engagement efficiently.

In addition to, the core principle of Fitts's law is "Smaller and farther targets take longer to reach" [1]. As a matter of fact of previous studies about Fitts's law that determine fundamental concepts of Fitts's law, how it was developed, and evaluating Fitts's law in user-friendly AR interfaces, VR/AR gaming design, and services applications to improve user satisfaction and engagement. In a similar way, when it comes to miller's law which targets "the average person can only keep 7 (plus two or minus two) in their working memory" [4]. Evaluating miller's law when the UI/UX design which consider as one of the best practices that designers can consider when building user interfaces.

Thus, the research focuses on implementing an AI-driven model for evaluating the Fitts's law, miller's law particularly the quantitative and measurable principles. Furthermore, this research leads to evaluate those principles through screen-based automation, computer vision and NLP algorithms. Not only that but also, it aims measure potential cognitive strain by quantitative evaluation metrics.

3. Statement of Research problem

UI/UX designers often face difficulties in evaluating whether their designs align with HCI principles due to lack of automation, limited knowledge of the principles, or the absence of integrated HCI laws support. As a result, they frequently need to update their UI based on user feedback, which can lead to designs that make users feel uncomfortable. There is a need clear for an AI-model that can evaluate quantitative HCI principles in real time to ensure high usability and accessibility from the early stages of design.

4. Aim and specific objectives

Aim:

The main aim of this research is to develop an AI-powered model that can evaluate quantitative HCI principles in UI/UX design to improve user satisfaction by enhancing the quality, usability, and accessibility of the UI design using Fitts's law and Miller's law.

Specific objectives;

1. To fast the development of lifecycles by proving early-stage feedback on UI design quality.
2. To implement an AI model capable of analyzing and detecting violations on UI design based on Fitts's Law and Miller's Law.
3. To generate visual or textual feedback for designers to support design improvements.

5. Research Design and Methods

1. Data Collection

   Collect UI screenshots (images) representing different Screen sizes, layouts, and UI styles. Aim for at least 100 samples per subcategory under Fitts's Law and Miller's Law. Sources which can be used,

   - Open-source design galleries
   - Real-world mobile applications
   - Design tools such as Figma

2. Data Preprocessing

   Image resizing: By using OpenCV, we have resized the all the UI design images into 375 X 750 pixels because according to the collected dataset resolutions, choose an average resolution to resize the images.

3. Annotation and Labelling Image

   Annotate and label each screenshot according to compliance with Fitts's Law and Miller's Law. Use image annotation tools such as Labelling, CVAT, Roboflow. Each UI element will be tagged to indicate size, spacing, and count of chunks to check whether it meets the respective HCI subcategories.

4. Extract Features

   Detect each UI element in the screenshot /image and extract key metrics related to each subcategory of Fitts's and Miller's Law.

   - **Fitts's Law**: -

- $Target\ Size = \{width\} \geq \{\min_size\} \wedge \{height\} \geq \{\min_size\}$

- $Distance = \sqrt{\{(x_2 - x_1)^2 + (y_2 - y_1)^2\}}$ (x, y coordinated values)

- $ID = log_2\left(1 + \frac{D}{S}\right)$ ID= Index of Difficulty, D= distance
  S= height or width from center of the screen

- $TP = \frac{ID}{MT}$ TP= Throughput, ID= Index of Difficulty
  MT= Movement Time (D/S)

- $Edge\ Advantage = \{Near\ edge\ if:\}x < \{margin\} \vee x > \{screen_width\} - \{margin\} \vee y < \{margin\} \vee y > \{screen_height\} - \{margin\}$

- **Miller's Law**: -

- $Cognitive\ Load = N_v + N_t$ Nv= No of Visible UI Elements
  Nt= total words

- $Chunking = \frac{G}{N_v}$ G= No of visually distinct grouped elements
  Nv= Total visual elements

- $SD = \frac{C}{S}$ SD= Step-wise Disclosure, C= total content chunks,
  S= No of progressive steps

- $WM = \sum_{\{i=1\}}^{n} w_i$ WM= Working Memory, W= weight or cost per element

- $NL = E \times M$ NL= Navigation Load, E= No of cross page elements,
  M= memory cost per transition

Use various tools and technologies such as,

- image processing: - Python, OpenCV, NumPy
- UI element detection: - YOLOv8, Detectron2, TensorFlow Object Detection API
- Feature Extraction: - Python logic for size, distance and OpenCV for count calculating.

5. Train AI model

Build an AI model to evaluate UI/UX Design Based on HCI Principles: Fitts's Law and Miller's Law in screenshots.

- Use CNN or transformer-based vision model for visual pattern recognition.
- Implement rule-based logic in Python script for subcategory-specific calculations.
- Finally train a Random Forest classifier to access pass/fail against each subcategory of laws.

6. Output and feedback
   Provide real-time feedback to user based on UI with a generator summary table showing which subcategories passed/failed.

7. Expected outcomes/alternative approaches

The key outcome of this research is an AI-powered model that evaluates UI/UX design using quantitative HCI principles, especially Fitts's Law and Miller's Law. The model will analyze whether each subcategory under these two laws is satisfied and provide real-time feedback based on the screenshot of the UI design uploaded by the designer or user. If a UI design fails to meet a law or one of its subcategories, the system will display the corresponding standards allowing designers to improve usability during early stages.

Expected Outcomes

- An AI-based prototype that capable of analyzing UI for compliance with Fitts's Law and Miller's Law subcategories.
- A real-time feedback (pass/fail) system for each subcategory.
- A set of standards derived from the model's outputs to guide early-stage design improvements.
- Summary report or table that shows the overall evaluation status of the UI design.
- A well-documented research report documenting methodology, model design and results.

Alternative Approaches/ Pitfalls

- If real-time processing effect delays, batch image processing can be used to ensure performance.
- It UI components are undetectable, enhanced image recognition techniques may be introduced.

8. Time Plan

Step	Time	Activity
Step 01 Data Collection	May 2024 - June 2025	• Collecting screen shots of mobile apps from external resources. • Analyze and collect relevant resources
Step 02 Data Preprocessing	July 2025 – August 2025	• Conducting data clean and labeling using tools to make user satisfied UI/UX
Step 03 Train the model	September - November 2025	• Developing AI models using appropriate algorithms
Step 04 Testing and Finalization	November – December 2025	• Compare pre and post-test to analyze the performance of the AI. • Get feedback.

Table 8.1: Time Plan

9. Individual Contribution

Index Number	Name	Contribution
TG/2020/673	M.G.H. Nirmani	Responsible for evaluating Fitts's Law
TG/2020/704	T. Madhusha	Responsible for evaluation Miller's Law

Table 9.1: Individual Contribution

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